

NEBLINA 1

Flight Dynamics Evidence Package

1. Executive Summary & Report Status

Evidence Package Generation Status

Subsystem	Status
Nominal Simulation	COMPLETE
Atmosphere Model	COMPLETE
Scenario Analysis	COMPLETE
Validation Data	COMPLETE
Monte Carlo Execution	COMPLETED

Flight Performance

KPI	Value
Apogee AGL	2605.1 m
Max Mach	0.67 M
Max Velocity	228.9 m/s
Max Total Acceleration	12.5 g
Max Dynamic Pressure (Max-Q)	28.3 kPa
Rail Exit Velocity	26.4 m/s
Flight Duration	112.7 s
Ground Drift	1123.5 m

Stability

KPI	Value
Static Margin (Rail Exit)	3.45 cal
Static Margin (Max-Q)	4.54 cal
Static Margin (Burnout)	4.66 cal
Max Angle of Attack (Ascent)	60.8 deg

Recovery

KPI	Value
Drogue Descent Rate	60.9 m/s
Main Descent Rate	8.9 m/s
Touchdown Velocity	8.8 m/s
Touchdown Energy	1135.0 J
Recovery Opening Shock	NOT PHYSICALLY RESOLVED BY CURRENT MODEL

Engineering Requirements Compliance

ID	Description	Result	Margin	Status
FD-LAUNCH-001	Vehicle must leave the launch rail with sufficient velocity for aerodynamic stability.	26.37 m/s	1.37	WITHIN GUIDELINE
FD-STABILITY-001	Static margin must remain strictly positive during the entire burn phase.	None	-	NOT EVALUATED
FD-STABILITY-002	Static margin must not exceed highly overstable limits causing extreme weathercocking.	None	-	NOT EVALUATED
FD-LOADS-001	Maximum airframe acceleration (thrust + aero) during ascent phase.	None	-	NOT EVALUATED
REC-LOADS-001	Maximum instantaneous opening shock deceleration.	None	-	NOT EVALUATED
REC-DESCENT-001	Orbiter parachute terminal descent rate.	None	-	NOT EVALUATED
REC-DESCENT-002	Main parachute terminal descent rate for safe kinetic energy ground impact.	None	-	NOT EVALUATED
REC-IMPACT-001	Total kinetic energy at touchdown.	None	-	NOT EVALUATED

2. Model Provenance & Input Quality

Physical Domain Validity Envelopes

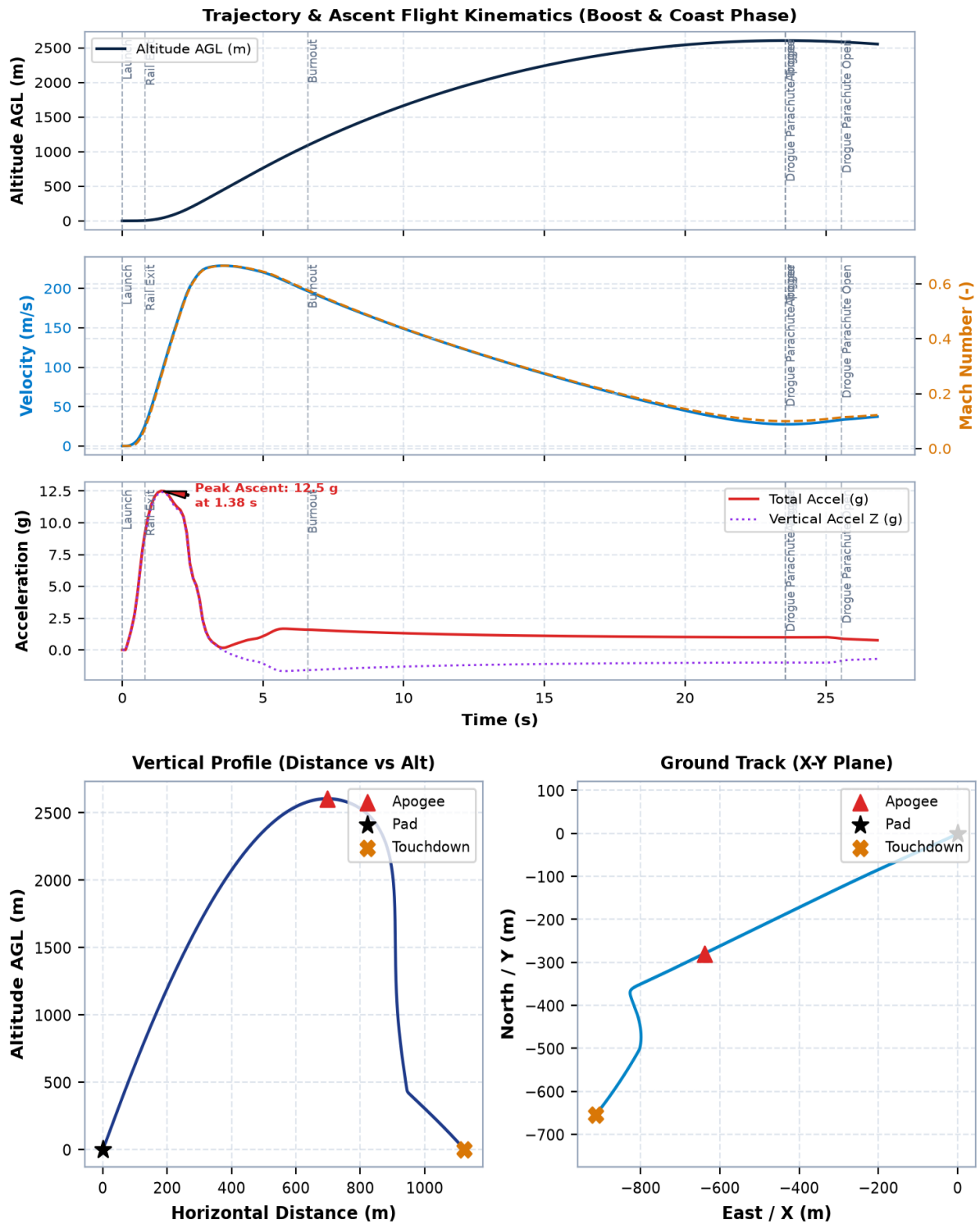
Domain & Parameter	Simulated Peak	Validity Envelope	Status	Note
Aerodynamics: Flight Mach Number	0.67 M	≤ 0.80 M (Subsonic)	QUALIFIED	Slender-body linear Barrowman equations valid without transonic wave drag corrections.
Aerodynamics: Ascent Angle of Attack	60.83°	$\leq 10.0^\circ$ (Linear Lift)	NON-LINEAR REGIME	Linear normal force slope CNa assumes attached flow without crossflow separation.
Atmosphere: Apogee Altitude ASL	3100 m	$\leq 30,000$ m (GFS Grid)	QUALIFIED	MAGI weather profile covers entire trajectory domain from ground pad to apogee.
Launch Dynamics: Rail Guide Clearance	26.4 m/s	≥ 25.0 m/s	QUALIFIED	Rail button release dynamics assume rigid tower alignment and aerodynamic stability.

Input Quality Classification

Parameter	Source	Classification	Confidence
Dry Vehicle Mass	Laboratory Precision Scale	MEASURED	HIGH (± 0.05 KG)
Center of Gravity (CG)	Dual Knife-Edge Balance Rig	MEASURED	HIGH (± 5 MM)
Moments of Inertia (Ixx, Iyy)	SolidWorks Full Assembly CAD Model	CAD ESTIMATE	MEDIUM ($\pm 8\%$)
Solid Motor Thrust Curve	Static Fire Test Bench Load-Cell	STATIC FIRE	HIGH ($\pm 2\%$ IMPULSE)
Propellant Initial Mass	Pre/Post Static Fire Weighing	MEASURED	HIGH (± 0.02 KG)
Aerodynamic Coeffs (Cd, CNa)	Barrowman Slender-Body Model	OPENROCKET / ROCKETPY	MEDIUM ($\pm 10\%$ CD)
Atmospheric Environment	MAGI Operational GFS Forecast	NUMERICAL WEATHER MODEL	ENSEMBLE (± 1.5 M/S WIND)

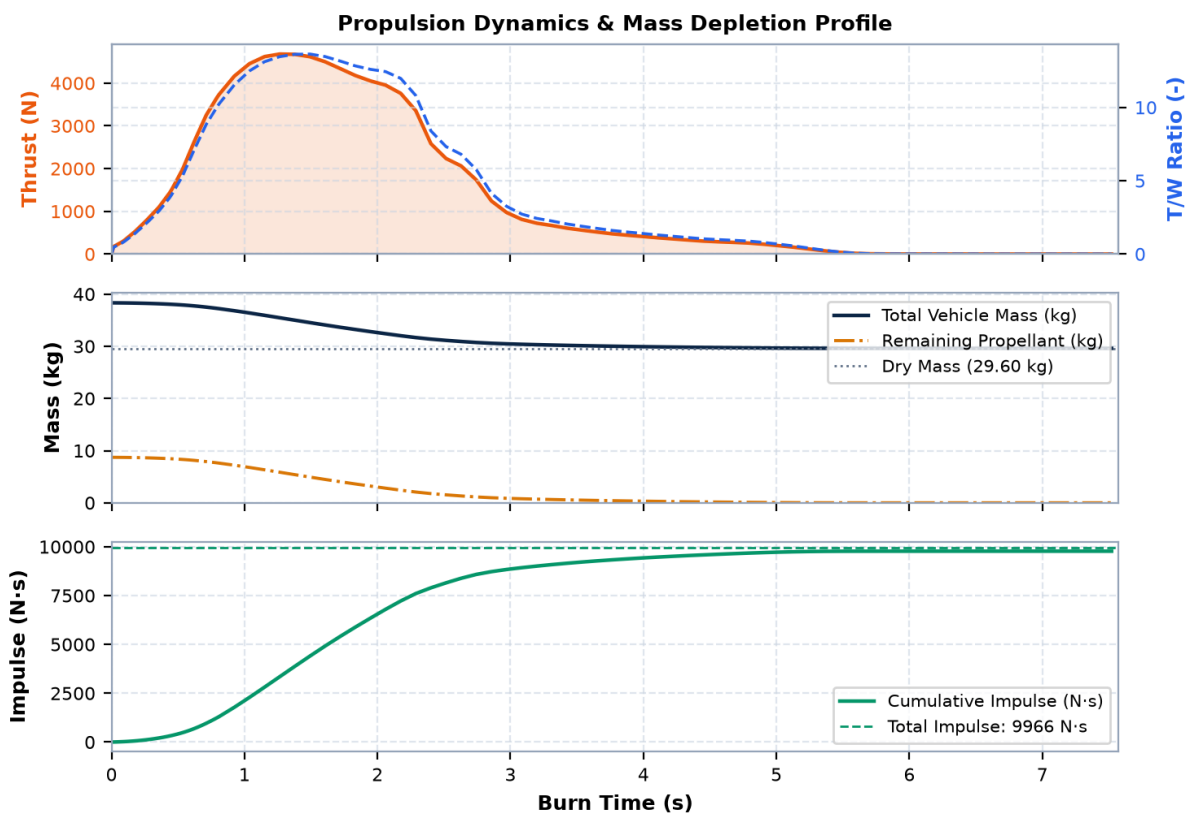
Parameter	Source	Classification	Confidence
Parachute Drag Areas (CdS)	Manufacturer Specification & Drop Test	ENGINEERING ESTIMATE	MEDIUM (±12% CDS)

3. Flight Kinematics & Trajectory



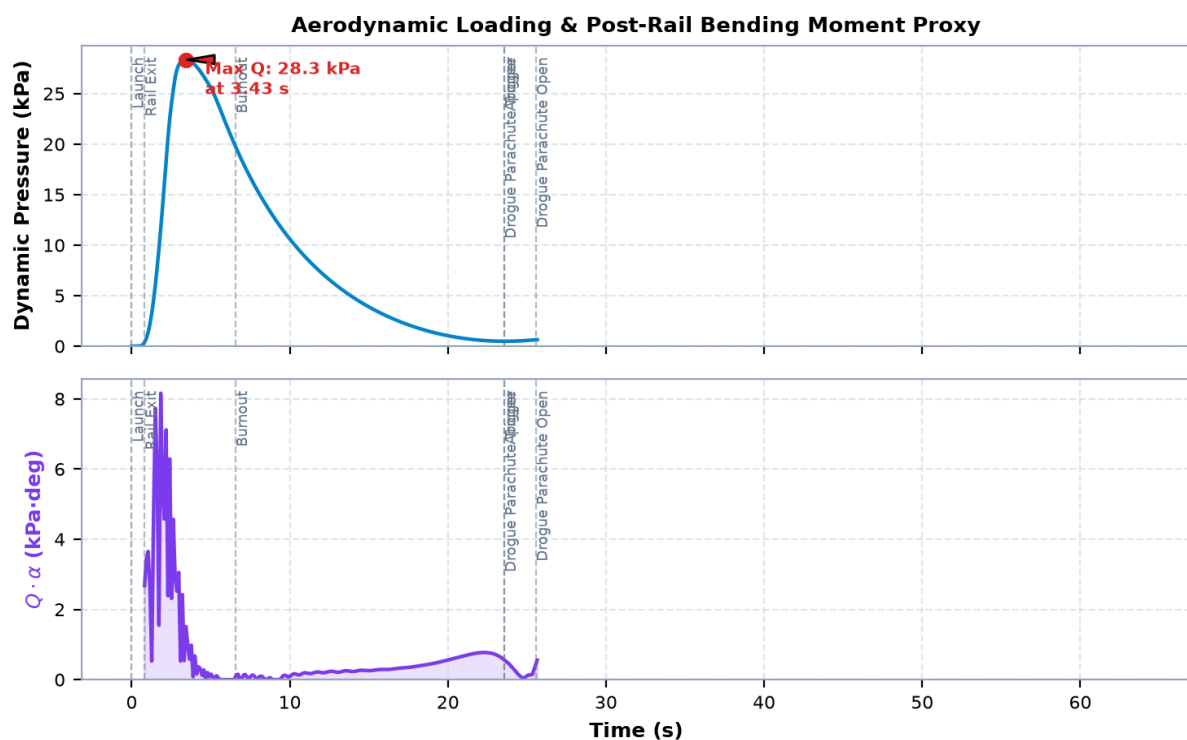
4. Mass & Propulsion Dynamics

Propulsion KPI	Value
Initial Mass	38.32 kg
Burnout Mass	29.60 kg
Propellant Mass	8.72 kg
Initial T/W	0.00
Peak T/W	13.64
Burn Time	6.59 s
Total Impulse	9966.4 Ns
Average Thrust	1513.3 N
Peak Thrust	4681.4 N



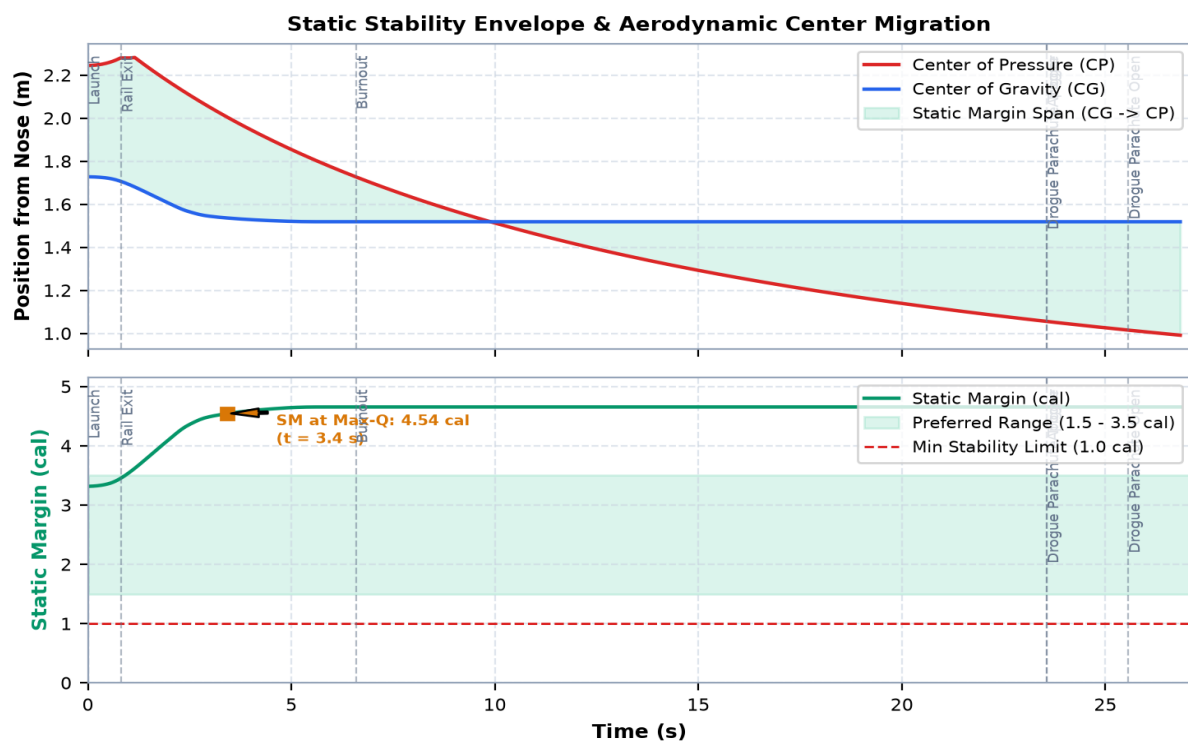
5. Aerodynamic Loads (Max Q & Bending)

Max-Q State	Value
Dynamic Pressure	28.3 kPa
Time	3.43 s
Altitude AGL	412.5 m
Mach Number	0.67 M
Velocity	226.1 m/s
Angle of Attack	0.1 deg
Static Margin	4.54 cal



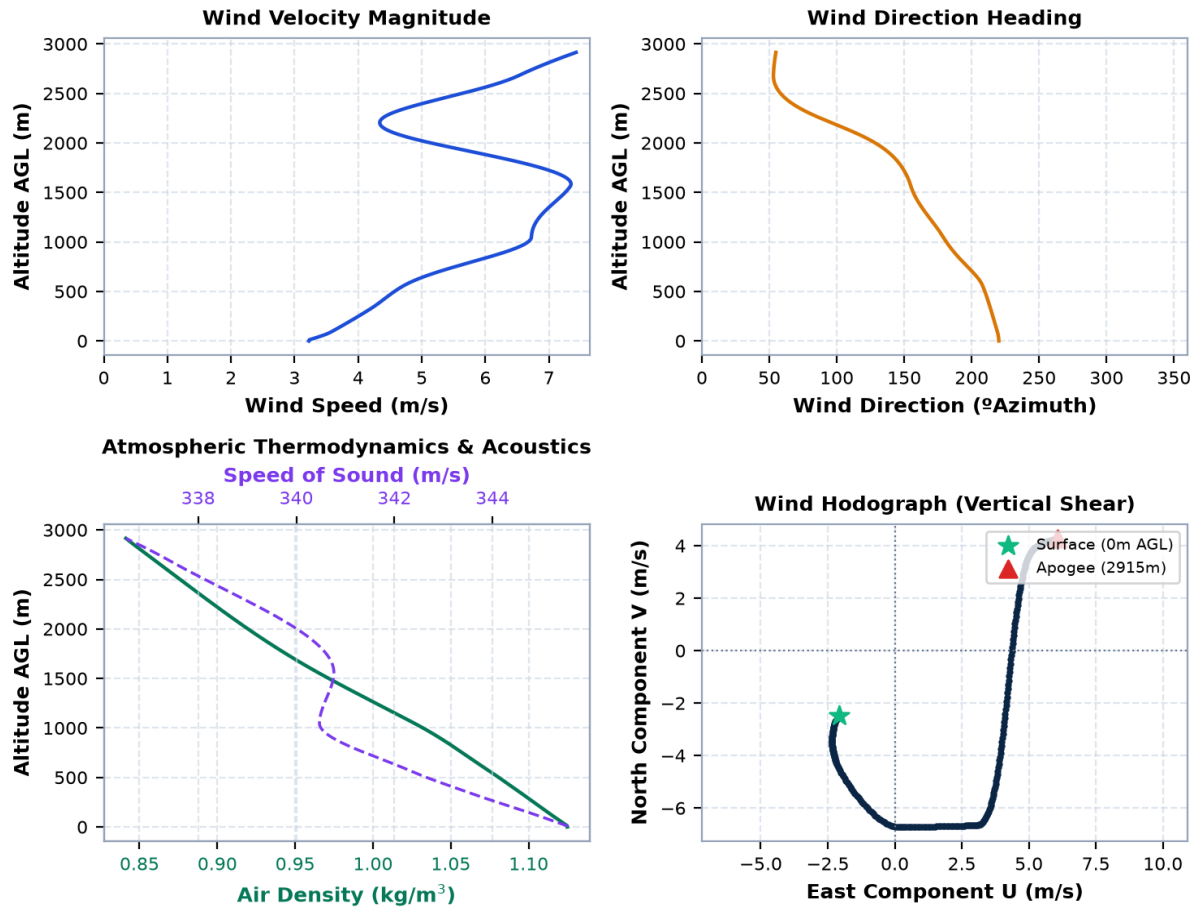
6. Stability Evolution

Stability Checkpoint	Margin (cal)
Liftoff	3.32
Rail Exit	3.45
Max-Q	4.54
Burnout	4.66
Maximum Margin	4.66

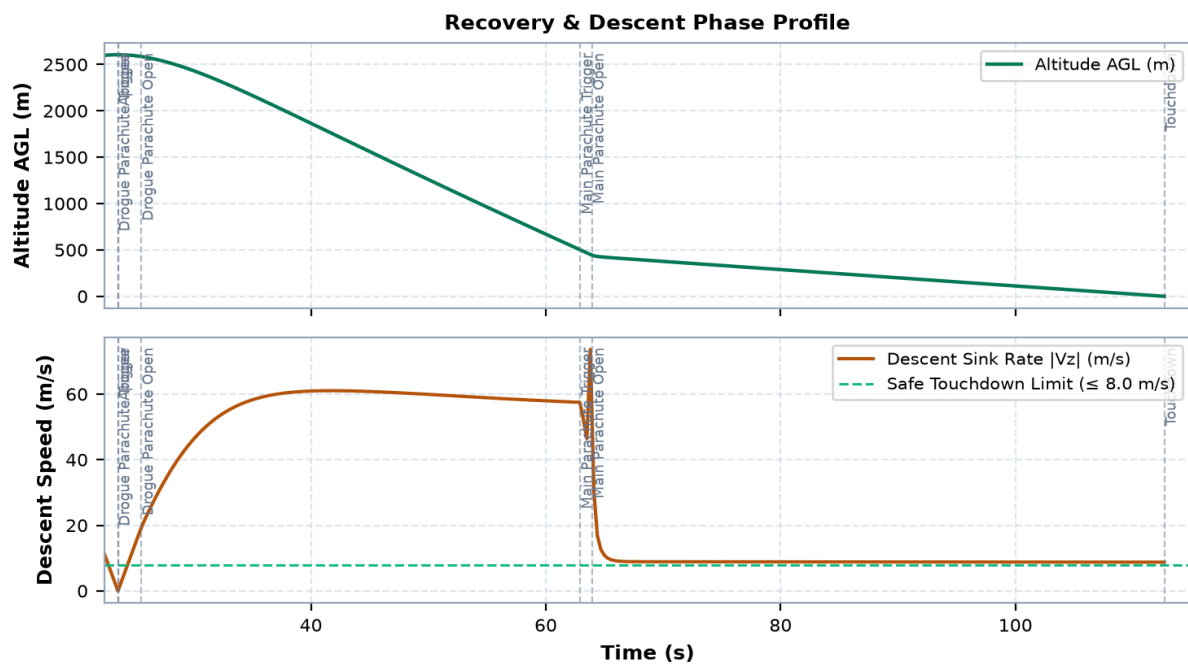


7. MAGI Atmospheric Profile

MAGI High-Fidelity Atmospheric Environmental Profile

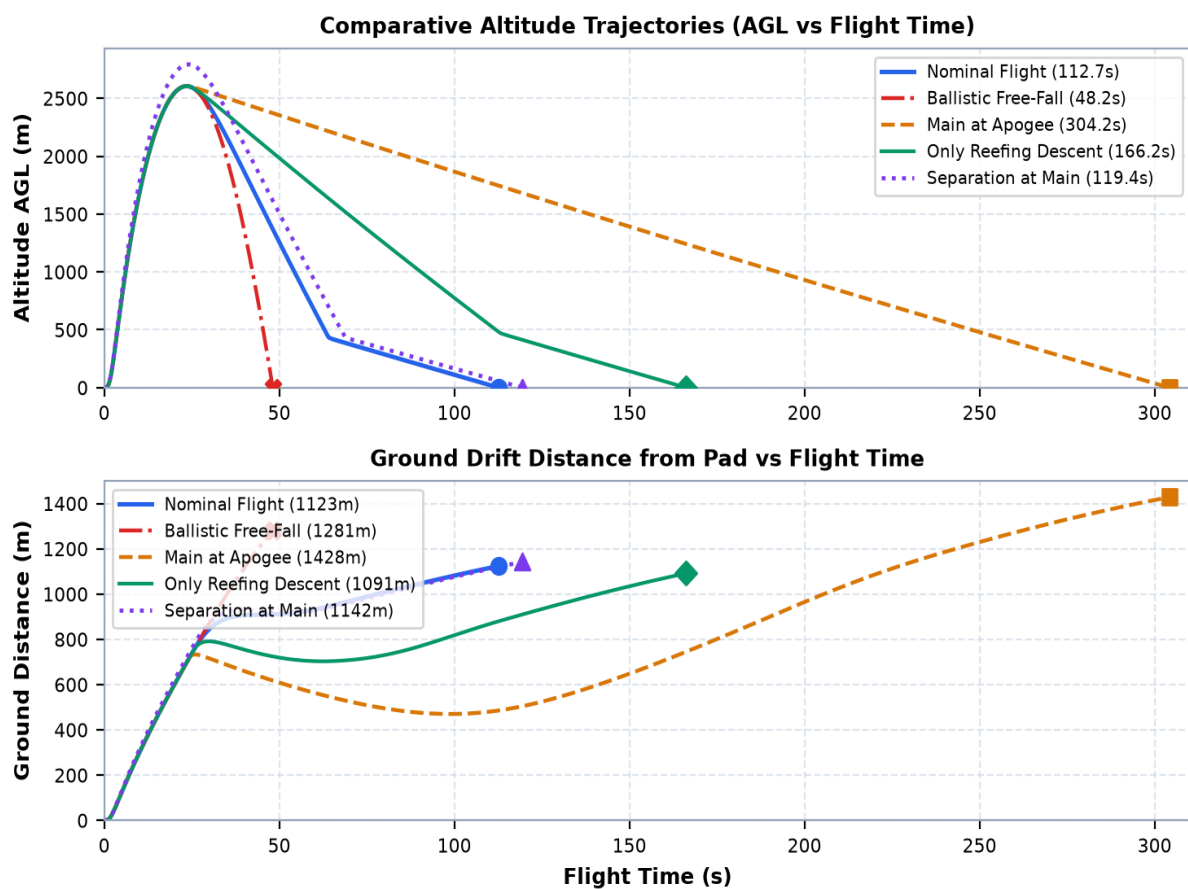


8. Recovery Profile



9. Multi-Scenario Comparative Flight

Scenario	Apogee (m)	Flight Time (s)	Max Mach	Max-Q (kPa)	Landing Dist (m)	Touchdown (m/s)
Nominal Flight	2605.1	112.7	0.67	28.3	1123.5	8.8
Ballistic Free-Fall	2605.1	48.2	0.67	28.3	1280.6	186.3
Main at Apogee	2605.1	304.2	0.67	28.3	1427.9	8.8
Only Reefing Descent	2605.1	166.2	0.67	28.3	1091.0	8.8
Separation at Main	2792.7	119.4	0.71	32.0	1142.1	8.5



10. Monte Carlo Stochastic Analysis

Stochastic Run Info	Value
Campaign ID	2026-09-17T035857Z_neblina_1_campaign
Status	COMPLETED
Requested Cases	20
Completed Cases	20
Failed Cases	0

Stochastic Inputs / Provenance

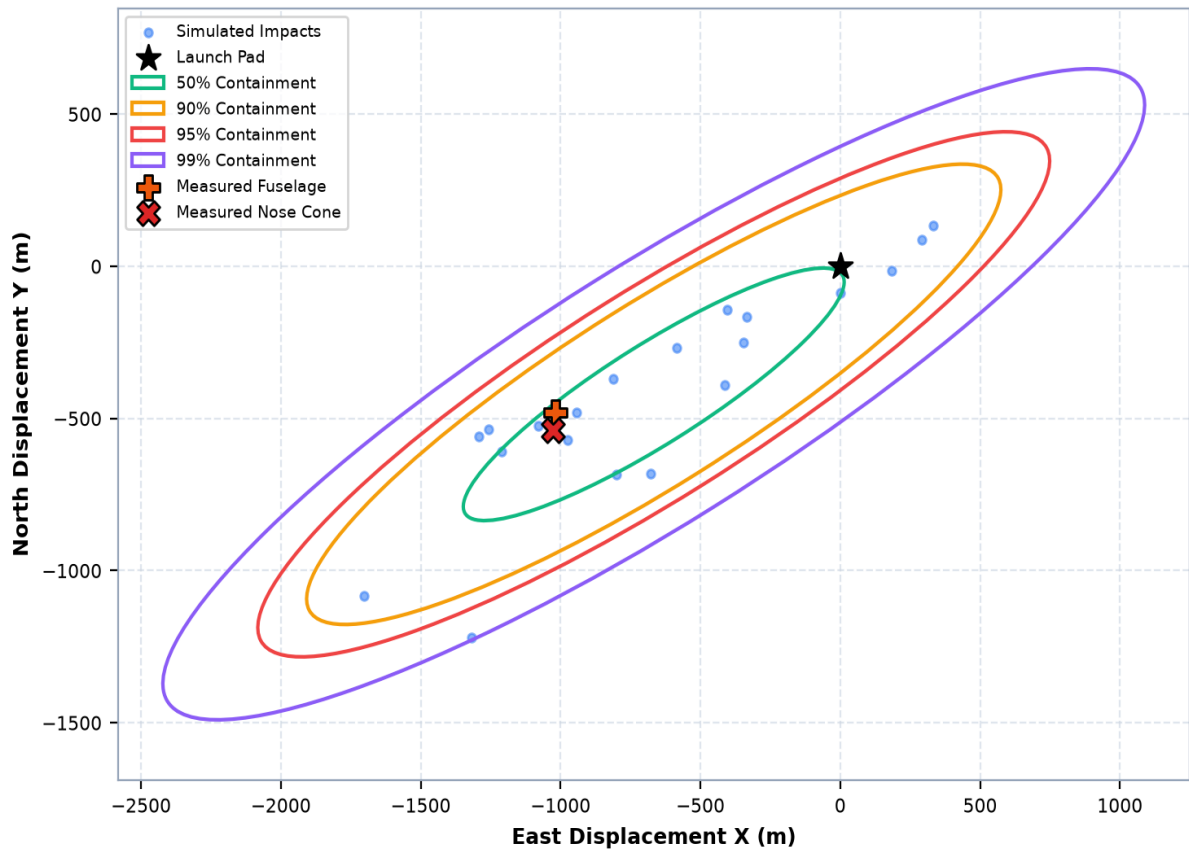
Parameter	Distribution	Uncertainty Limits	Units	Status
launch_inclination	uniform	U[70.0, 90.0]	deg	LEGACY_ASSUMPTION
launch_heading	uniform	U[230.0, 250.0]	deg	LEGACY_ASSUMPTION
power_off_drag_multiplier	normal	N(1.0, 0.100)	dimensionless	LEGACY_ASSUMPTION
power_on_drag_multiplier	normal	N(1.0, 0.100)	dimensionless	LEGACY_ASSUMPTION
vehicle_mass	normal	N(1.0, 0.010)	kg	LEGACY_ASSUMPTION
total_impulse_multiplier	normal	N(1.0, 0.050)	dimensionless	LEGACY_ASSUMPTION
atmosphere_ensemble	categorical	Ensemble	categorical	NATIVE_MODEL_ENSEMBLE
site_elevation	normal	N(1.0, 5.000)	m	LEGACY_ASSUMPTION

Output Distributions

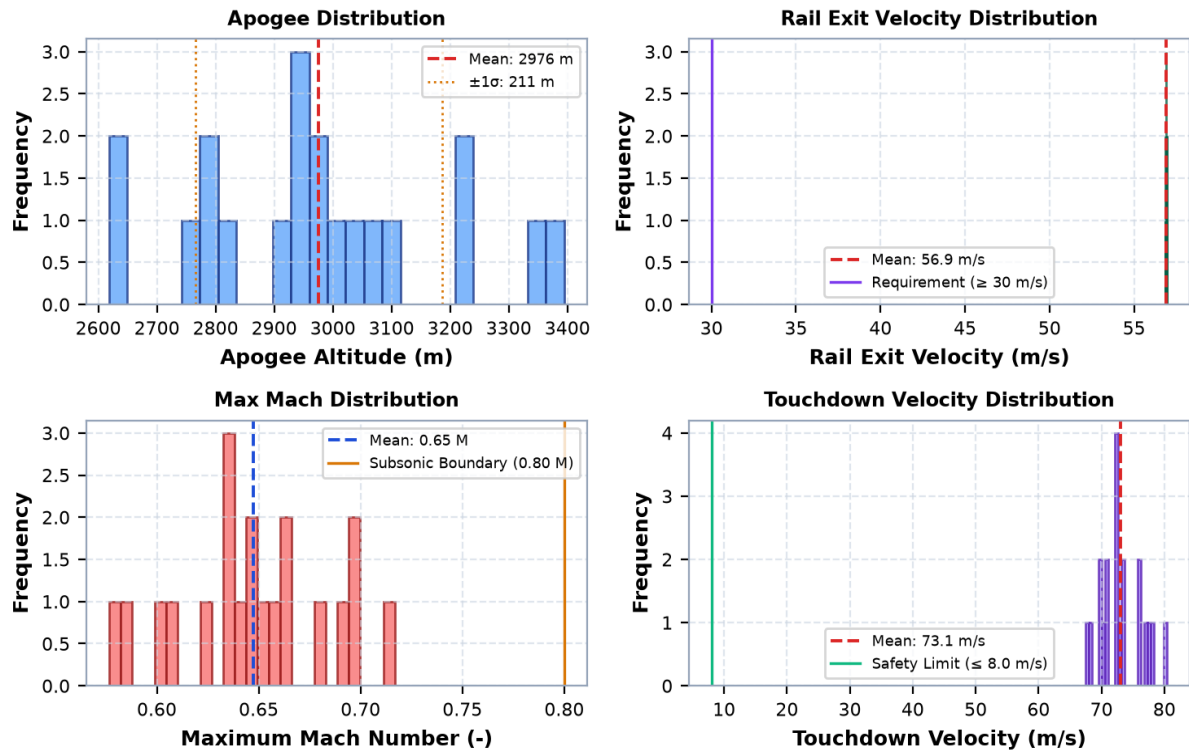
Variable	Mean	Std	P05	P50	P95
Apogee AGL (m)	2976.10	205.20	2629.24	2960.19	3336.42
Rail Exit Vel (m/s)	56.89	0.04	56.83	56.90	56.94
Max Mach (M)	0.65	0.04	0.58	0.64	0.70
Max-Q (kPa)	26.92	2.97	21.91	26.61	31.20
Landing X (m)	-666.66	563.69	-1338.40	-738.41	294.12
Landing Y (m)	-420.66	343.54	-1090.48	-434.95	90.42
Touchdown (m/s)	-73.05	3.40	-78.13	-72.44	-68.13
Flight Time (s)	68.87	4.15	62.09	68.35	76.00

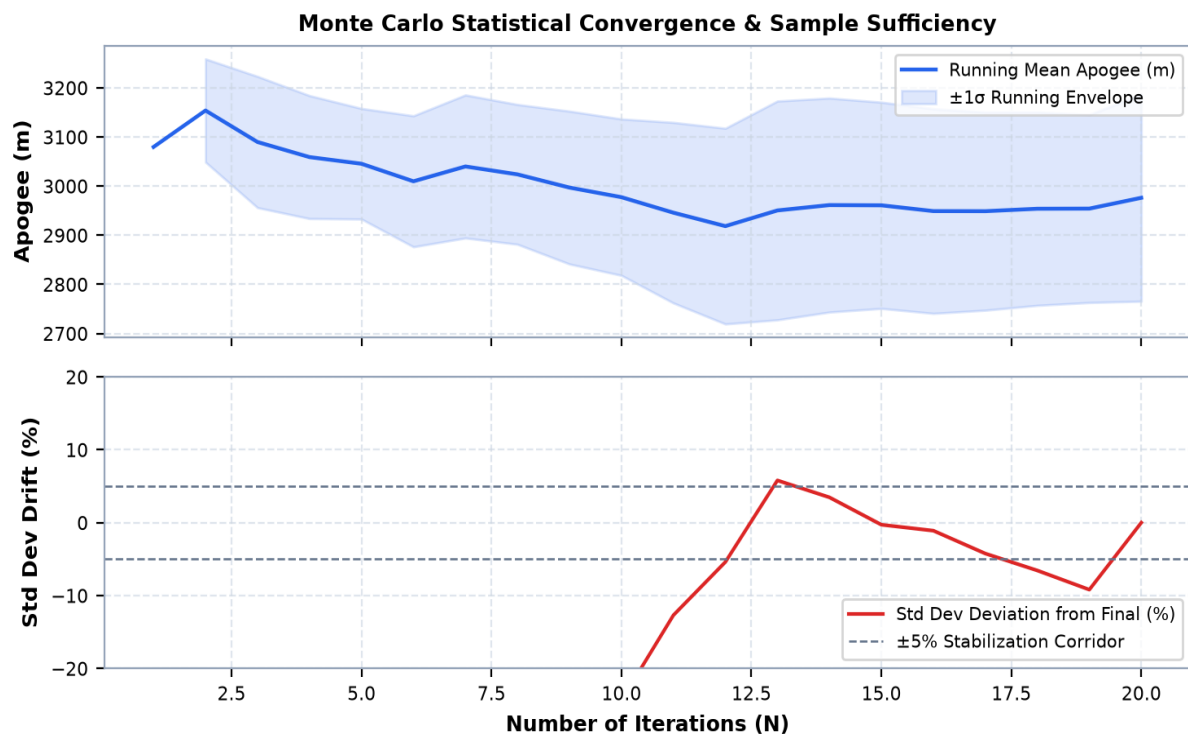
Stochastic Requirement Confidence Matrix

Monte Carlo 2D Ground Dispersion & Containment Ellipses (20 Cases)



Monte Carlo Output Distributions & Requirement Compliance





Reproducibility Appendix

Attribute	Value
Execution Timestamp (UTC)	2026-09-17T03:59:31.113188Z
Git State	5bf285ed9ec5e9c9703e9f4f1ee6914f13fe0ebb (DIRTY)
Project Directory	C:\Users\Arthur\PycharmProjects\Flight_Dynamics_Antares\projects\neblina_1
Antares FD Framework	Version 2.0 (Hardened)

This report was automatically synthesized by Antares Flight Dynamics. No manual edits were performed on these charts or metrics. All values are traceable to the defined inputs, meteorological sources, and aerodynamic models.